

Cognitive Systems: Neural Networks and Applied AI

Exercise 4

Learning with Spiking Neural Networks

Javier López Randulfe, Etienne Müller, Nico Reeb

1. Recap biological neural networks

Javier López Randulfe, M.Sc.

Three generations of neuron models

1st generation

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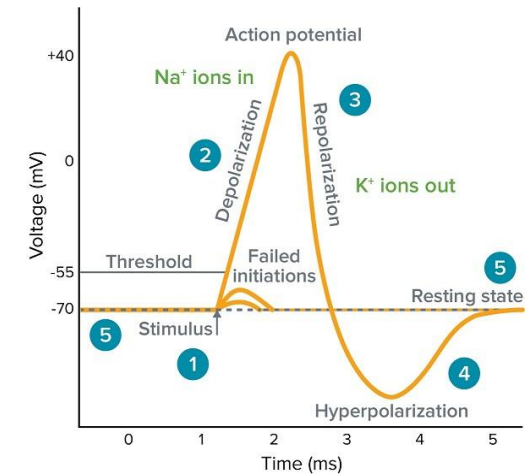
      1      1      0
    0      0      1
  1      0      1      1
    1      1      0
  
```

2nd generation

```

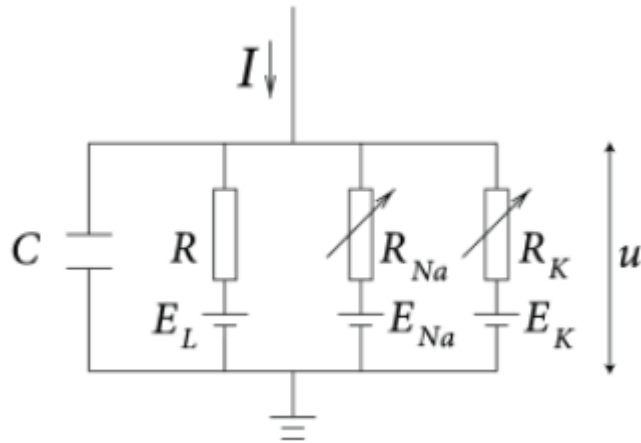
0.234 -0.645 0.764
      0.937 0.322
0.854 -0.544
      -0.208
  
```

3rd generation



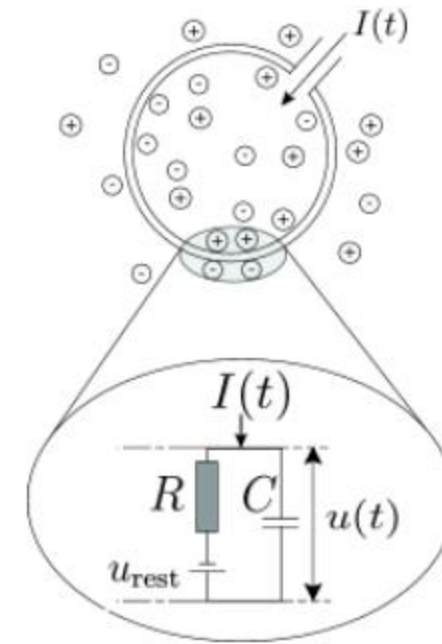
Neuron firing models

Hodgkin-Huxley



W. Gerstner, Neuronal Dynamics, 2014

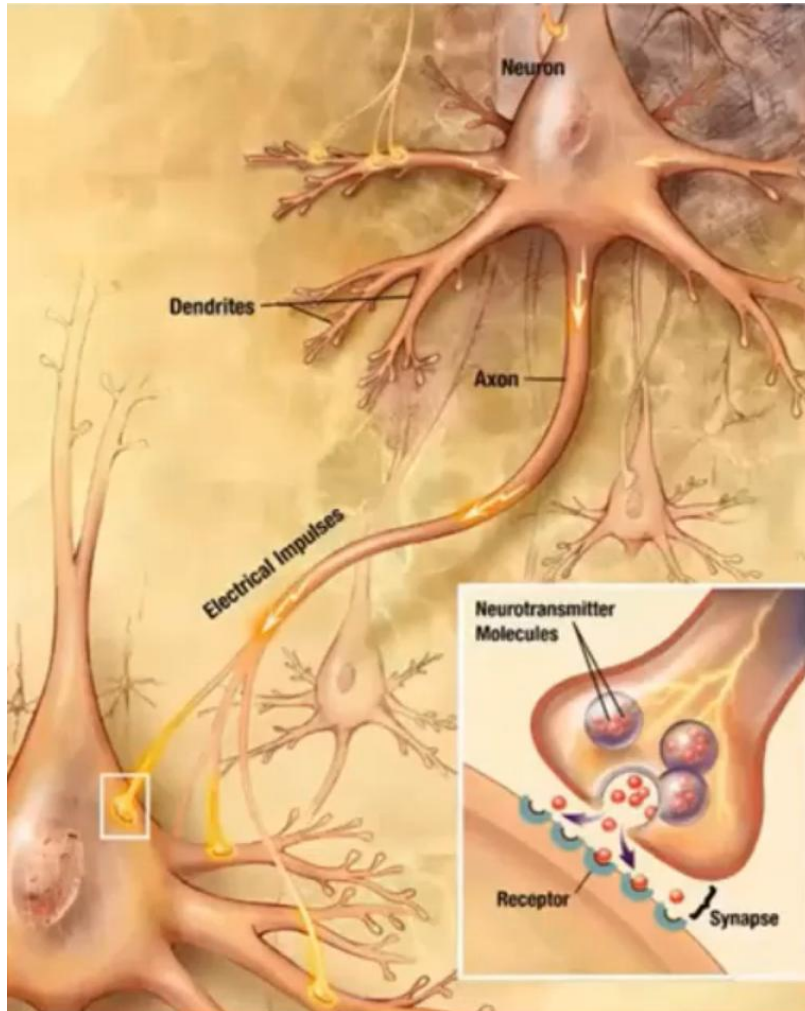
Leaky Integrate-and-Fire



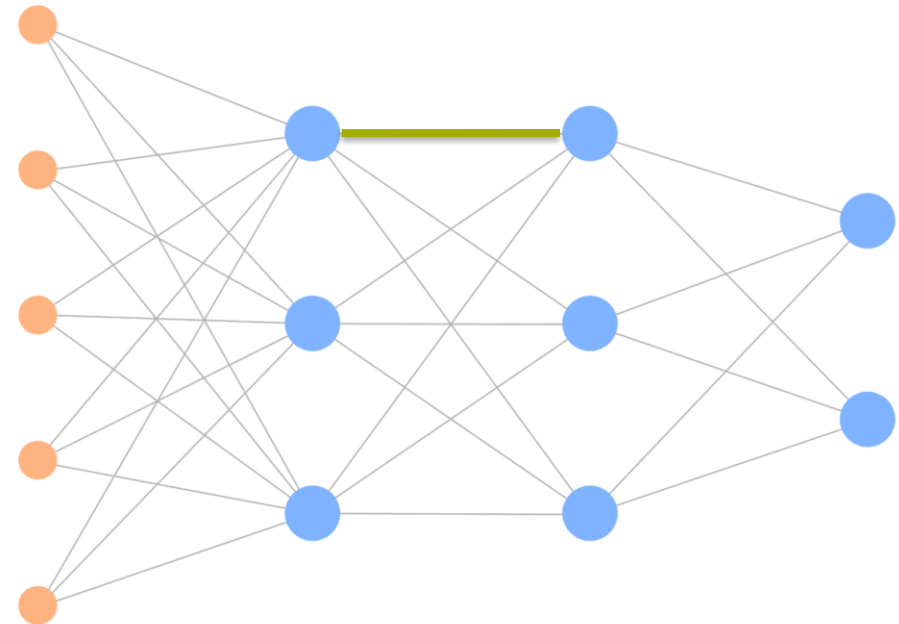
2. Learning

Javier López Randulfe, M.Sc.

From neurons to neural networks

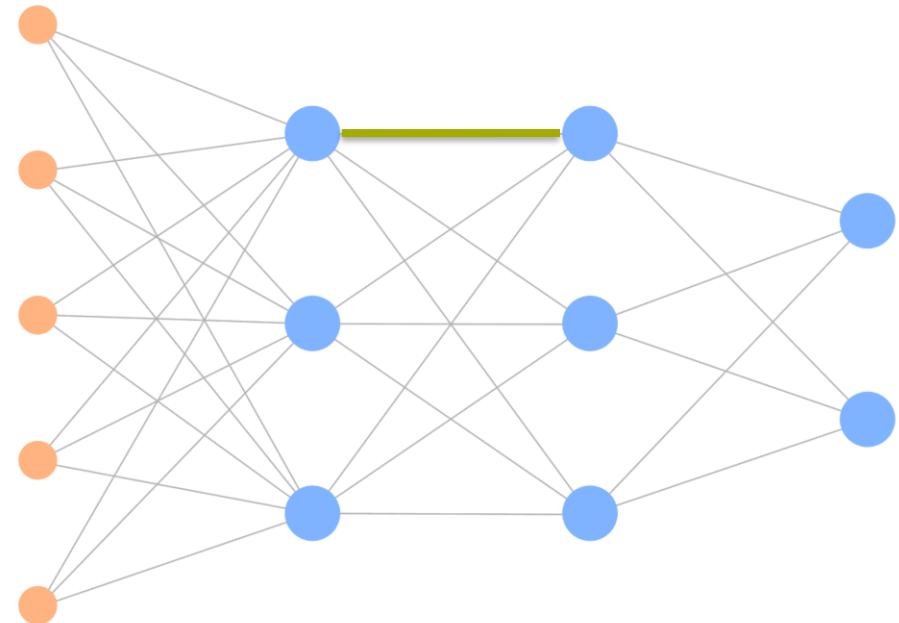


Source: Wikimedia Commons



What is learning?

- Learning = finding the right weights to solve a task
- **Universal approximation theorem:** a neural network with only a single hidden layer can approximate any function arbitrarily well.



Classification of Learning Models

Based on the error feedback:

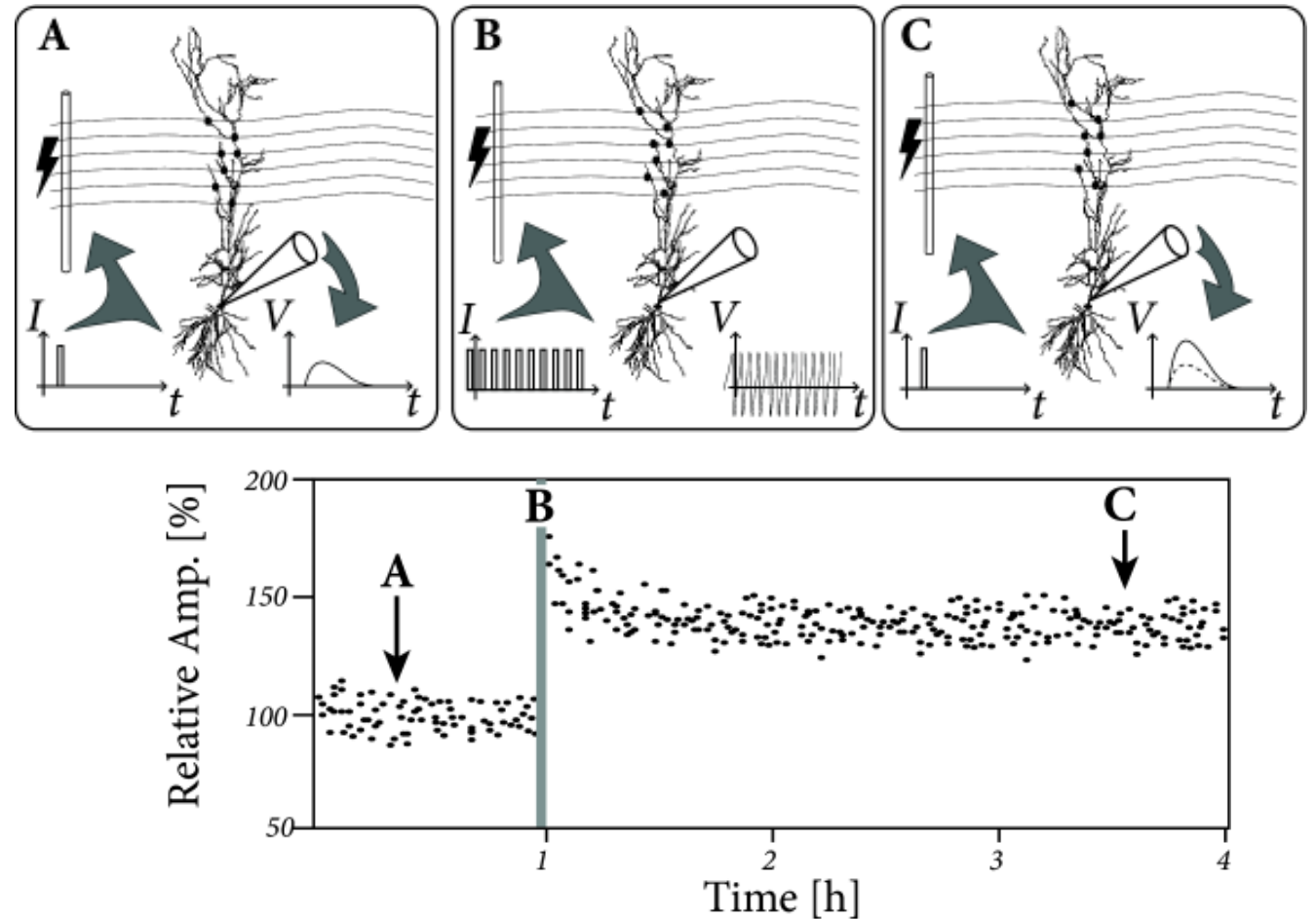
- Supervised learning
 - Unsupervised learning → **Hebbian learning**
 - Reinforcement learning
- 
- Rate-based hebbian rules
 - Precise-timing-based hebbian rules

Based on the model precision:

- **Phenomenological** → black-box model
- Biophysical → internal electro-chemical processes

Hebbian learning

- "When an axon of cell A is near enough to excite cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased."*
Hebb, 1949
- Neurons that *"fire together, wire together"*
Shatz, 1992

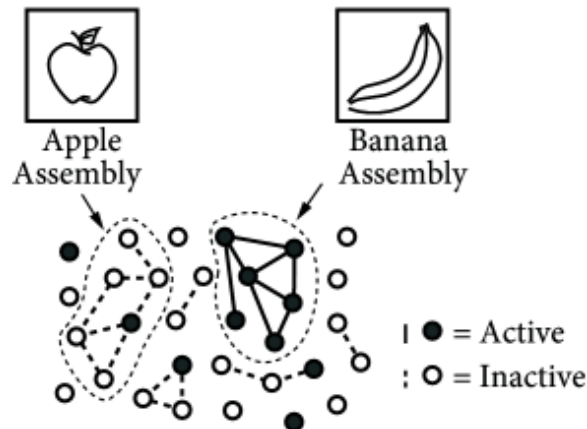


W. Gerstner, Neuronal Dynamics, 2014

Hebbian learning

How does it behave?

Long term potentiation (LTP) and long term depression (LTD)



How to model Hebbian learning?

- We may assume that the change of weights depends on the activity of the pre- and post-synaptic neuron, and the weight itself:

$$v_i = \vec{w} \cdot \vec{u}$$

$$\frac{dw_{ij}}{dt} = f(w_{ij}, v_i, u_j)$$

$v_i \equiv$ Post-synaptic neuron firing rate

$u_j \equiv$ Pre-synaptic neuron firing rate

$w_{ij} \equiv$ Synaptic weight

- Taylor approximation (see exercises)

Hebbian learning

Fundamental properties

- **Saturation**

Avoid unbound growth of synaptic weights, e.g., $0 < w < w_{max}$

- **Competition → Selectivity**

Avoid weights to converge to the same value, e.g., $\sum_j w_{ij}^2 = 1$ (normalization)

- **Locality**

Weight changes depend on local variables
(Biologically unfeasible otherwise)

Dayan, P., & Abbott, L. F. (2001). *Theoretical neuroscience: computational and mathematical modeling of neural systems*.

From Basic Hebb's Rule to Covariance Rule

Basic Hebb's rule

$$\tau_w \frac{d\vec{w}}{dt} = v\vec{u}$$

Correlation-based rule

$$\tau_w \frac{d\vec{w}}{dt} = \langle v\vec{u} \rangle = Q \cdot \vec{w}$$

Covariance-based rule

$$\tau_w \frac{d\vec{w}}{dt} = \langle v(\vec{u} - \theta_u) \rangle = C \cdot \vec{w}$$

$$\theta_u = \langle u \rangle$$

Introduces **LTD** (If $u < \theta_u$)

v $\equiv$ post-synaptic firing rate
 $\vec{u}$ $\equiv$ pre-synaptic firing rates
 τ_w $\equiv$ learning time constant
 Q $\equiv$ correlation matrix
 C $\equiv$ input covariance matrix
 $\langle \rangle$ $\equiv$ average operator

- ✗ Saturation
- ✗ Competition
- ✓ Locality

Dayan, P., & Abbott, L. F. (2001). *Theoretical neuroscience: computational and mathematical modeling of neural systems*.

Oja's Rule

Principal Components Analysis (PCA)*

$$\frac{d\vec{w}}{dt} = v\vec{u} - \alpha v^2 \vec{w}$$

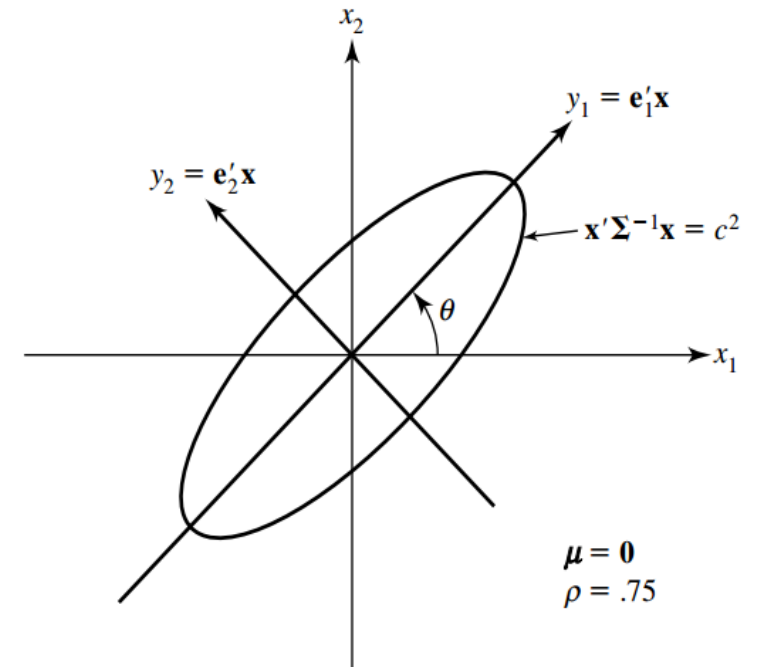


$$\frac{d\vec{w}}{dt} = C\vec{w} - (\vec{w}^T C \vec{w}) \vec{w}$$

$C \equiv$ input covariance matrix = $(\vec{u} \cdot \vec{u})$

$\alpha \equiv$ learning rate

- ✓ Saturation
- ✓ Competition
- ✓ Locality



Dayan, P., & Abbott, L. F. (2001). *Theoretical neuroscience: computational and mathematical modeling of neural systems*.

*Recommended bibliography on PCA: *Johnson, R. A., & Wichern, D. W. (2002). Applied multivariate statistical analysis. Chapter 8*

Bienenstock-Cooper-Munro (BCM) Theory

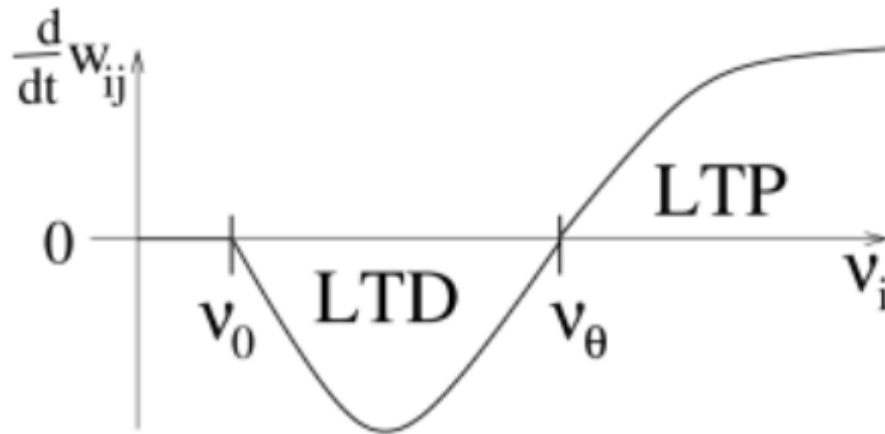
Covariance learning with sliding threshold

$$\frac{dw_{ij}}{dt} = \phi(v_i - v_\theta)u_j$$

$\phi(\cdot) \equiv$ non-linear function

Threshold rate v_θ depends on the average activity of the post-synaptic neuron:

$$v_\theta = f(\langle v_i \rangle)$$



- ✓ Saturation
- ✓ Competition
- ✓ Locality

W. Gerstner, Neuronal Dynamics, 2014

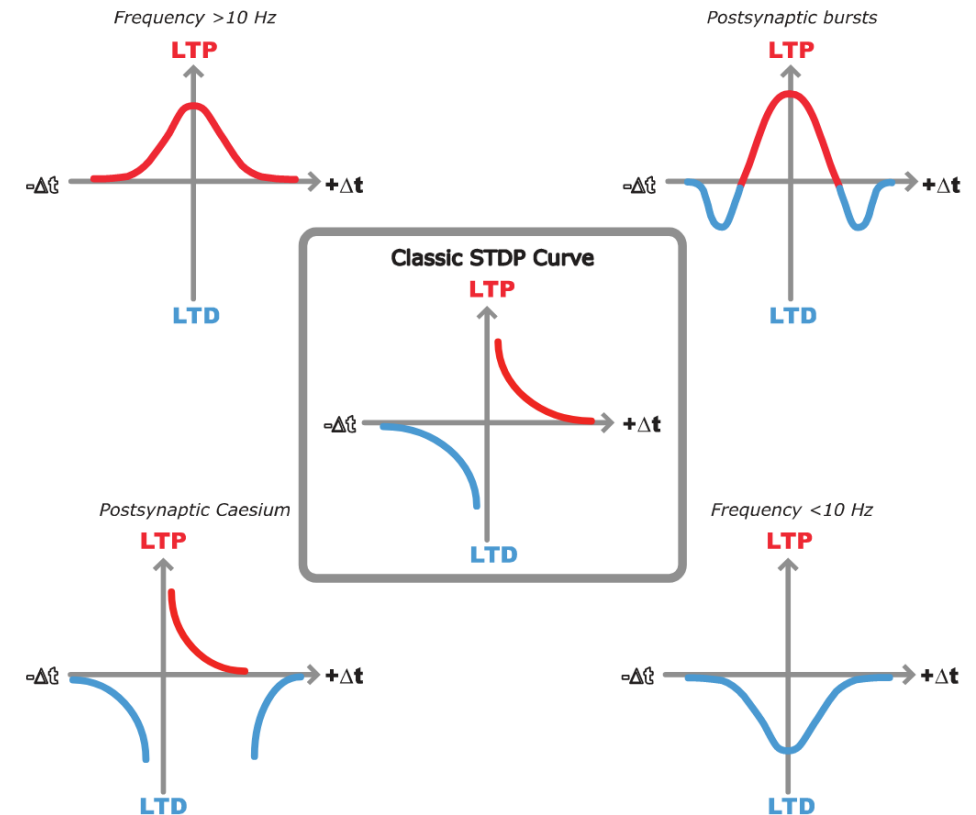
Dayan, P., & Abbott, L. F. (2001). *Theoretical neuroscience: computational and mathematical modeling of neural systems*.

Spike-timing dependent plasticity (STDP)

General Information

- *"Spike Timing Dependent Plasticity is a temporally asymmetric form of Hebbian learning induced by tight temporal correlations between the spikes of pre- and postsynaptic neurons."*
- The weight changes depends on the difference of the firing times of two connected neurons

Modelling the time dependence



Buchanan & Mellor (2010)



2. Learning - Exercises

Javier López Randulfe, M.Sc.

Two exercises – see additional PDF for details

Hebbian learning and Oja's rule

- Explore the Taylor expansion of Hebbian learning
- Observe problems in the time evolution of the weights for simple rules
- Derive Oja's rule
- Code implementation of Oja's rule

Making sense of differential STDP updates

- Consider how STDP could be realized in a biological system
- Show that the differential approach to STDP is equivalent to the approach shown in the slides
- Discover a computationally more efficient algorithm for updating the weights

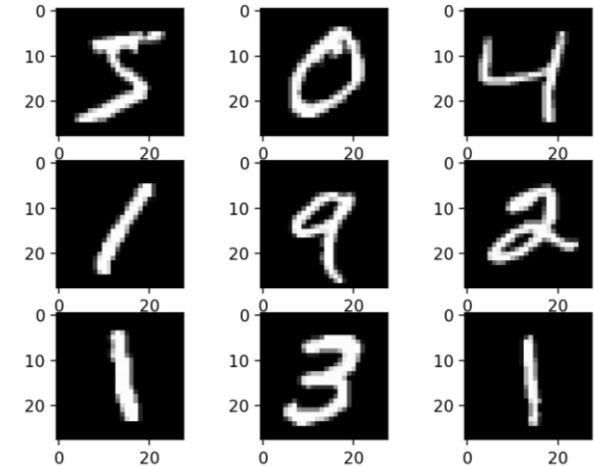
3. Conversion of ANNs to SNNs

Etienne Müller, M.Sc.

SNN Benchmarks on MNIST

Single Hidden Layer Feedforward Network

- ANN with Backpropagation¹: 98.6%
- SNN with STPD²: 95%



Convolutional Neural Network:

- ANN with Backpropagation: 99.8% (~achievable Limit)
- SNN with STDP³: 98.4%

→ Backpropagation is the most powerful learning algorithm to date!

¹ McDonnal et al. (2015) Fast, Simple and Accurate Handwritten Digit Classification by Training Shallow Neural Network Classifiers with the 'Extreme Learning Machine' Algorithm

² Diehl and Cook (2015) Unsupervised learning of digit recognition using spike-timing-dependent plasticity. Frontiers in Computational Neuroscience, 9, 1–9.

³ Tavaneh and Madeia (2016) Bio-inspired spiking convolutional neural net- work using layer-wise sparse coding and STDP learning, arXiv preprint arXiv: 1611.03000, 1–16.

Backpropagation in SNN (SpikeProp)

Spike-events are non-differentiable!

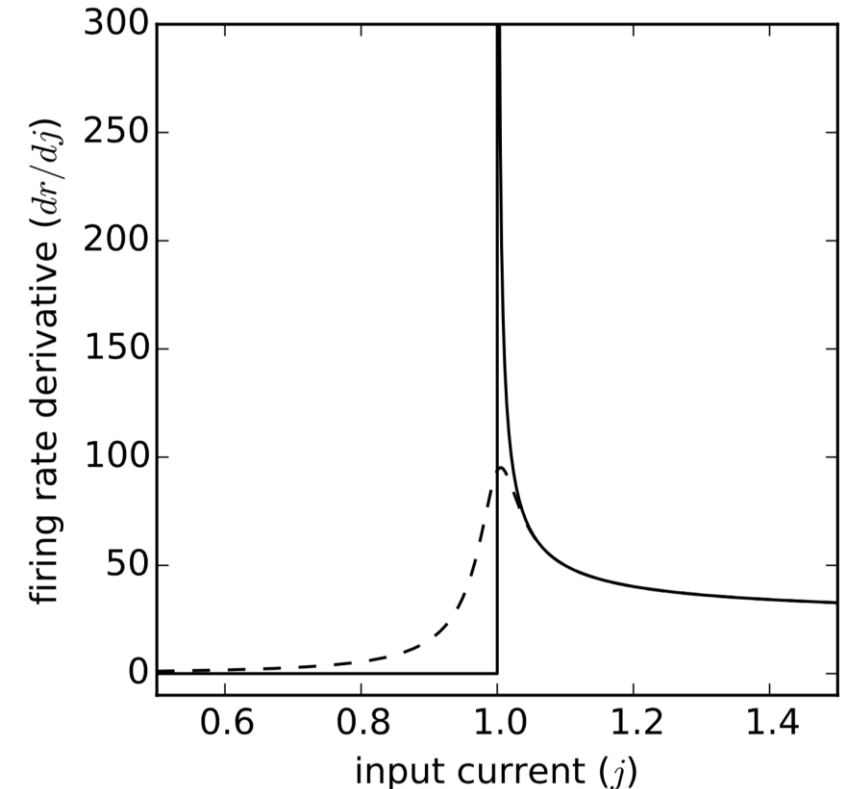
Different approaches for enabling spiking backpropagation:

- Treating discontinuities as noise¹
- Differentiable approximation²

→ Better learning, worse performing neurons

¹ Lee, J. H., Delbruck, T., & Pfeiffer, M. (2016). Training deep spiking neural networks using backpropagation

² Hunsberger et al. (2016) Training Spiking Deep Networks for Neuromorphic Hardware



SNN Benchmarks on MNIST

Feedforward Network

- ANN (SLFN) with Backpropagation: 98.6%
- SNN (SLFN) with STPD: 95%
- **SNN (2 hidden layer) with Backpropagation¹: 97.2%**

Convolutional Neural Network:

- ANN with Backpropagation: 99.8% (~achievable Limit)
- SNN with STDP: 98.4%
- **Spiking CNN with Backpropagation²: 99.3%**

¹ Tavanaei, A., Kirby, Z., & Maida, A. S. (2018a). Training spiking convnets by STDP and Gradient Descent

² Lee, J. H., Delbruck, T., & Pfeiffer, M. (2016). Training deep spiking neural networks using backpropagation

How to use the full potential?

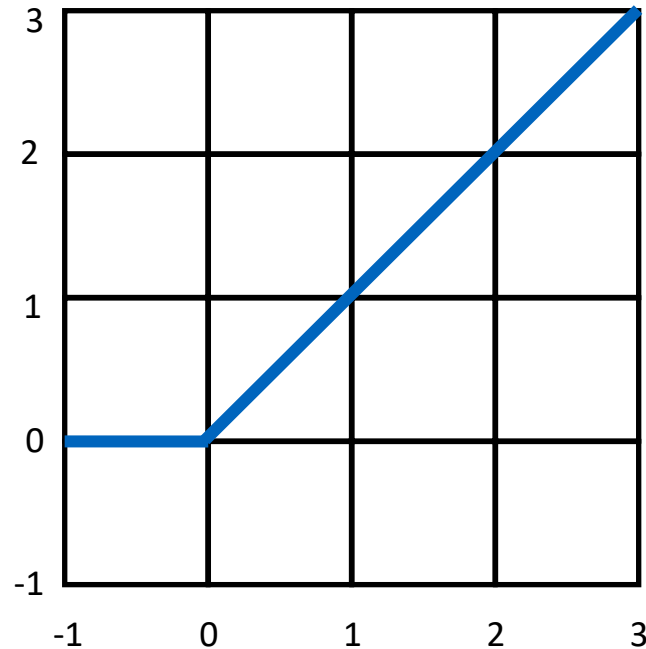
Train ANN

- Design and train an analogue neural network
- Highly optimized algorithms and large variety of available deep learning libraries will result in great networks

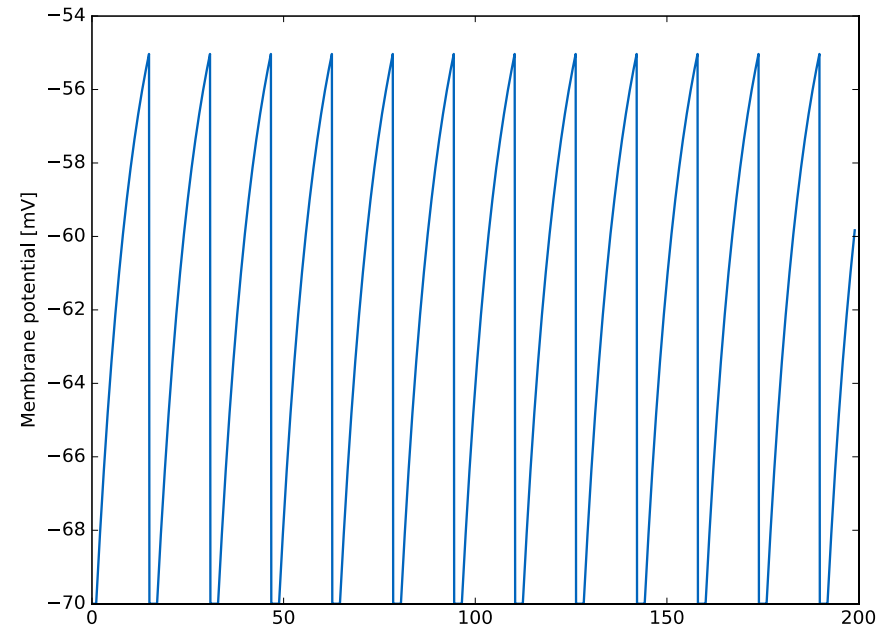
Convert to SNN

- Simply replace the activation function with a spiking neuron model

ReLU vs. Rate Coding



vs.



Problem:

Spiking neurons have a maximum frequency, ReLU has no upper Limit

→ would result in high conversion loss

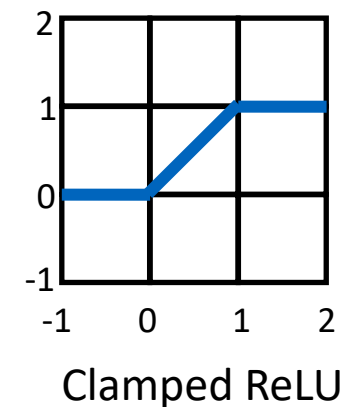
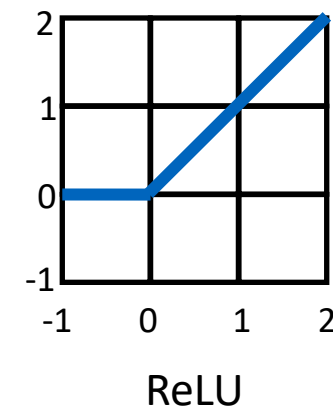
Fix the Upper Limit Problem

Normalization

- Calculate the max activation of the ReLU neurons
- Normalize the weights of the network by that value

Constrain-then-train

- Limit the network before training for ideal conversion performance
- Use a clamped ReLU



Conversion of Feedforward Networks

Simply replace the ReLU activation function of the normalized network (of the constrained trained network, respectively) with a spiking neuron

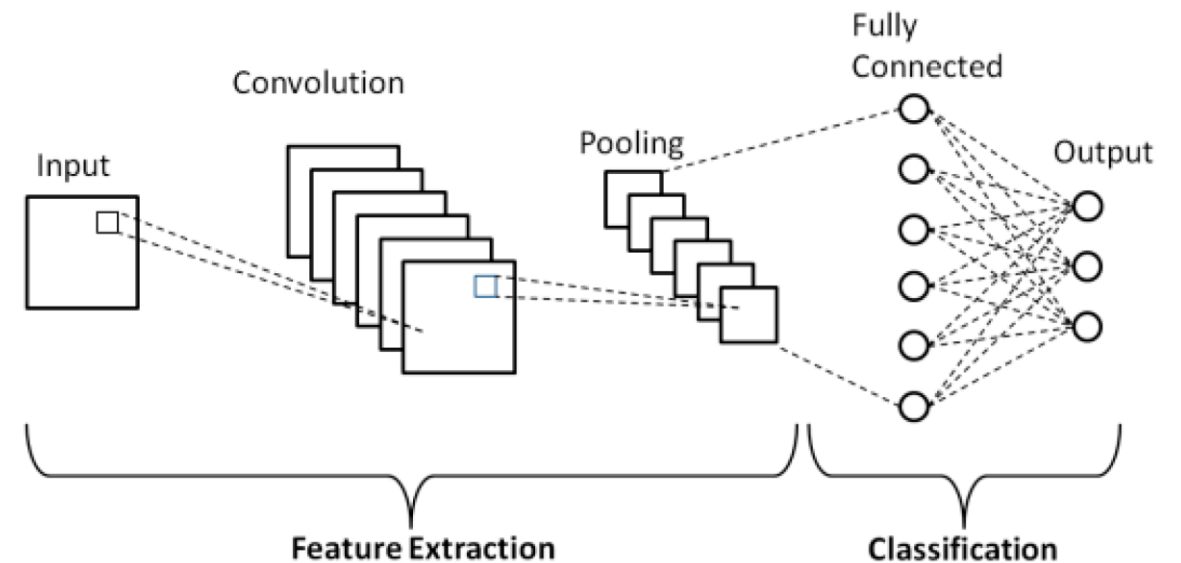
→ Conversion near-lossless

Conversion of Convolutional Neural Networks

Features that can't be directly translated into SNN:

- Max pooling
- Softmax
- Batch normalization

→ Methods or alternatives have been proposed to achieve near-lossless conversion.



SNN Benchmarks on MNIST

Feedforward Network

- ANN (SLFN) with Backprop: 98.6%
- SNN (SLFN) with STPD: 95%
- SNN (2 hidden layer) with Backprop: 97.2%
- **ANN (2 hidden layer) with Backprop, converted¹: 99.4%**

Convolutional Neural Network:

- ANN with Backpropagation: 99.8% (~achievable Limit)
- SNN with STDP: 98.4%
- Spiking CNN with Backpropagation: 99.3%
- **ANN with Backpropagation, converted²: 99.5%**

¹ Esser et al. (2015). Backpropagation for energy-efficient neuromorphic computing. In Advances in neural

² Rueckauer et al. (2017). Conversion of continuous-valued deep networks to efficient event-driven networks for image classification

Conversion of Recurrent Neural Networks

- Only little research done in this field!
- Conversion of Elman network: Accuracy dropped from 85% to 74%¹
- Conversion of LSTMs and GRUs still open
→ Research at TUM

¹ Diehl et al. (2016) Conversion of artificial recurrent neural networks to spiking neural networks for low-power neuromorphic hardware

But why?

Why should we use SNNs, if ANNs give us better results?

→ next week

4. Learning over years

- Synaptic plasticity, is that all?

Nico Reeb, M.Sc.

Neuroplasticity

“... the ability to make adaptive changes related to the structure and function of the nervous system”

Neuronal plasticity as an adaptive property of the central nervous system. **Karl Zilles (1992)**

Functional plasticity

- alter and adapt the functional properties of neurons

Structural plasticity

- physical creation and deletion of synapses

Activity-dependent structural plasticity. **Butz, Wörgötter and van Ooyen (2009)**

Experience-dependent structural plasticity in the adult brain: How the learning brain grows. **Schmidt, et al. (2020)**

Epigenetic changes and innate structures

“evidence for the role of epigenetic modifications [...] in both the stability and plasticity of developing neuronal circuits“

Epigenetic influences on brain development and plasticity.
Fagiolini, Jensen and Champagne (2009)

→ **innate genetic programs** leading to the sculpting of neuronal circuits during early brain development

“ evidence that the ability to use information about an adult’s direction of gaze and emotional expression to predict action is both present, and developing at the end of the first year of life“

Infants' ability to connect gaze and emotional expression to intentional action. Phillips, Wellman and Spelke (2002)

→ cognitive systems possess **innate structures** i.e. born with basic innate concepts:

- Motion detection
- Tracking
- Hand detection
- Gaze following

Epigenetic changes and innate structures

The brain structure has been evolving over millenia, in order to learn and adapt to the environment

- **Neurons** and basic neural circuits started developing around 600 to 700 million years ago
- **Basic brain structure** developed more than 500 million years ago e.g. brainstem, thalamus, or basal ganglia
- **Neocortex** has been under continuous development for the last 120 million years
- Current human **brain structure** developed 35.000 to 100.000 years ago

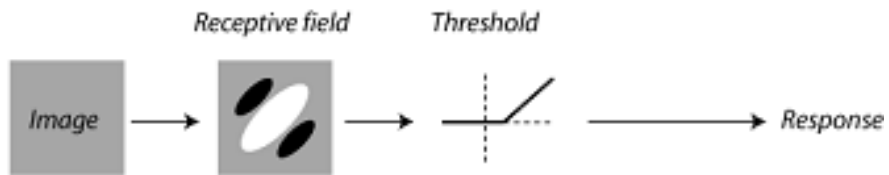
5. Sensory information and learning

- Making sense of neural circuits

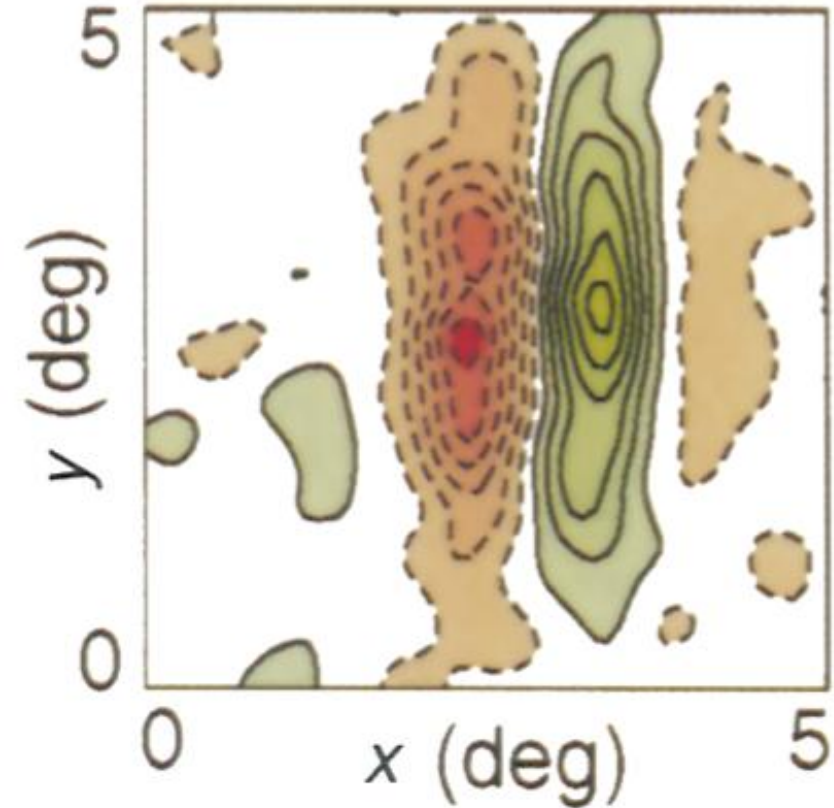
Nico Reeb, M.Sc.

Primary Visual Area (V1) – Simple cells

A Simple cell



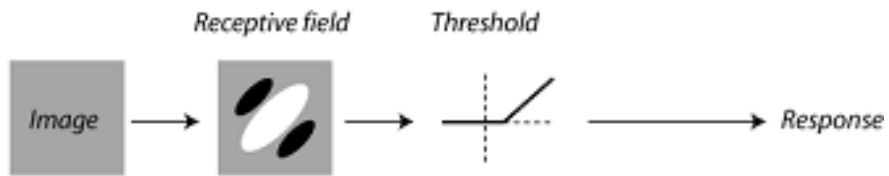
What simple and complex cells compute. **Carandini (2006)**



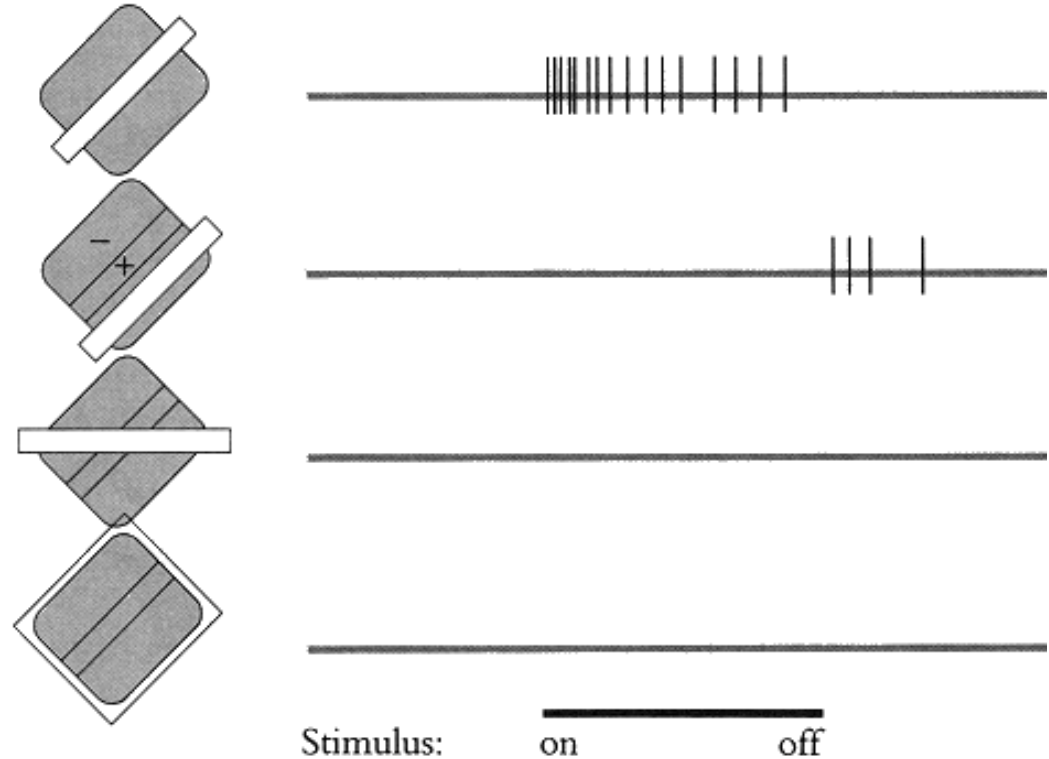
Receptive-field dynamics in the central visual pathways.
DeAngelis et al. (1995)

Primary Visual Area (V1) – Simple cells

A Simple cell



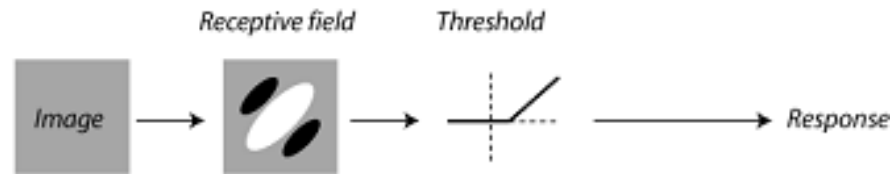
What simple and complex cells compute. **Carandini (2006)**



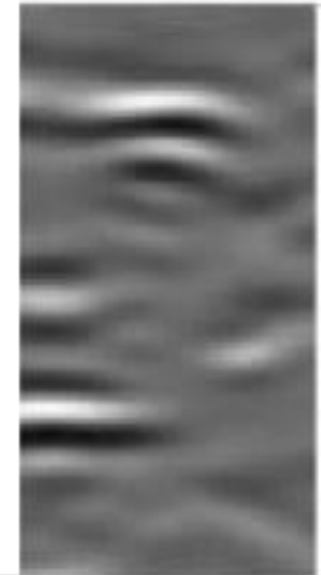
Eye, Brain and Vision. **Hubel (1995)**

Primary Visual Area (V1) – Simple cells

A Simple cell



What simple and complex cells compute. **Carandini (2006)**



Neural Basis of Pattern Vision. Kiper and **Carandini (2003)**

Primary Visual Area (V1) – Simple cells

Independent Component Analysis (ICA)

seperates **statistically independent** and **non-Gaussian** signals

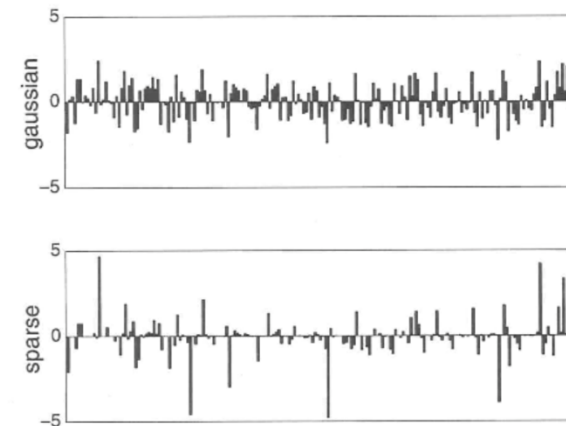


$$p(s_1, s_2) = p(s_1)p(s_2)$$

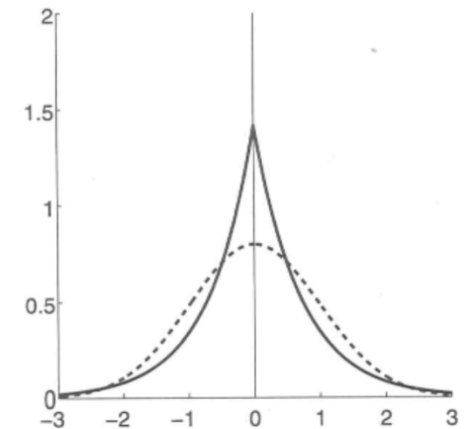


e.g. sparse distributions

A



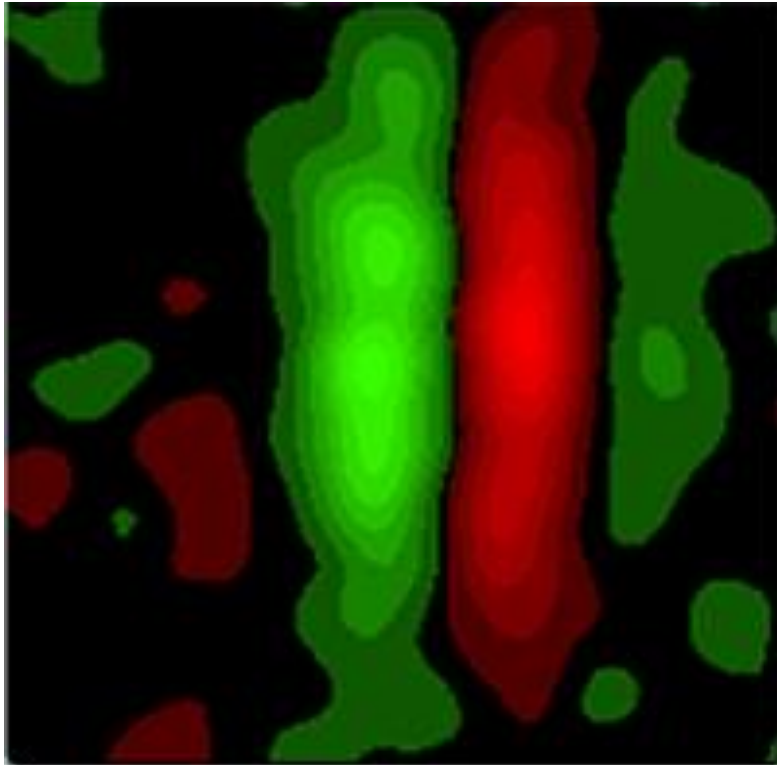
B



Natural Image Statistics. Hyvärinen, et al. (2009)

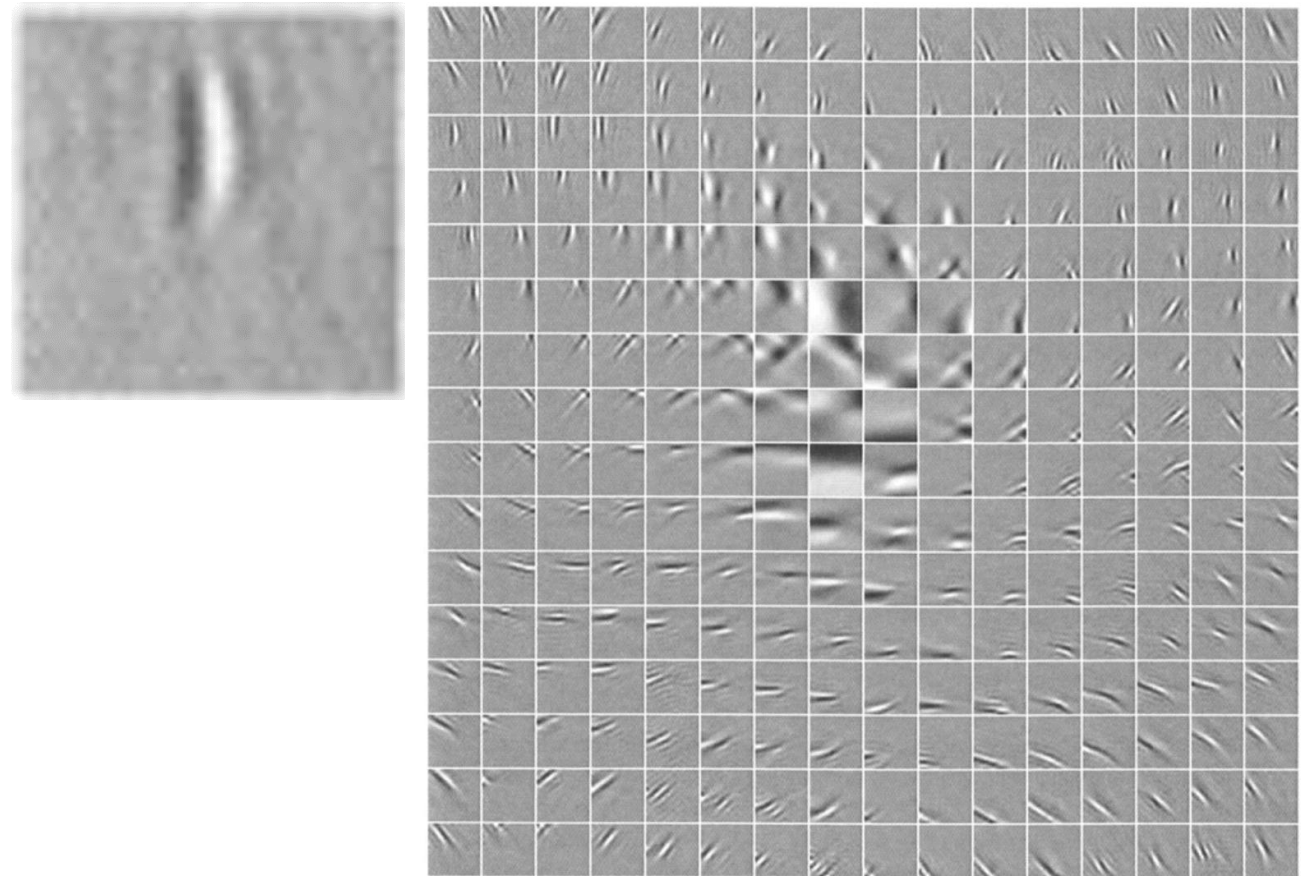
Primary Visual Area (V1) – Simple cells

Receptive field of simple cell



Receptive-field dynamics in the central visual pathways.
DeAngelis et al. (1995)

Filter set of ICA applied to natural scenes



Natural Image Statistics. **Hyvärinen, et al. (2009)**

Sensory Cue Integration

letters to nature

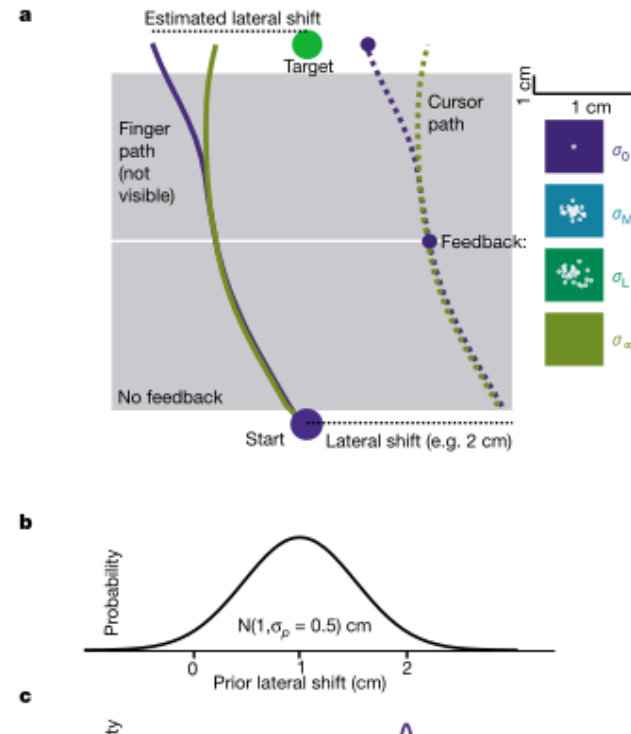
Bayesian integration in sensorimotor learning

Konrad P. Körding & Daniel M. Wolpert

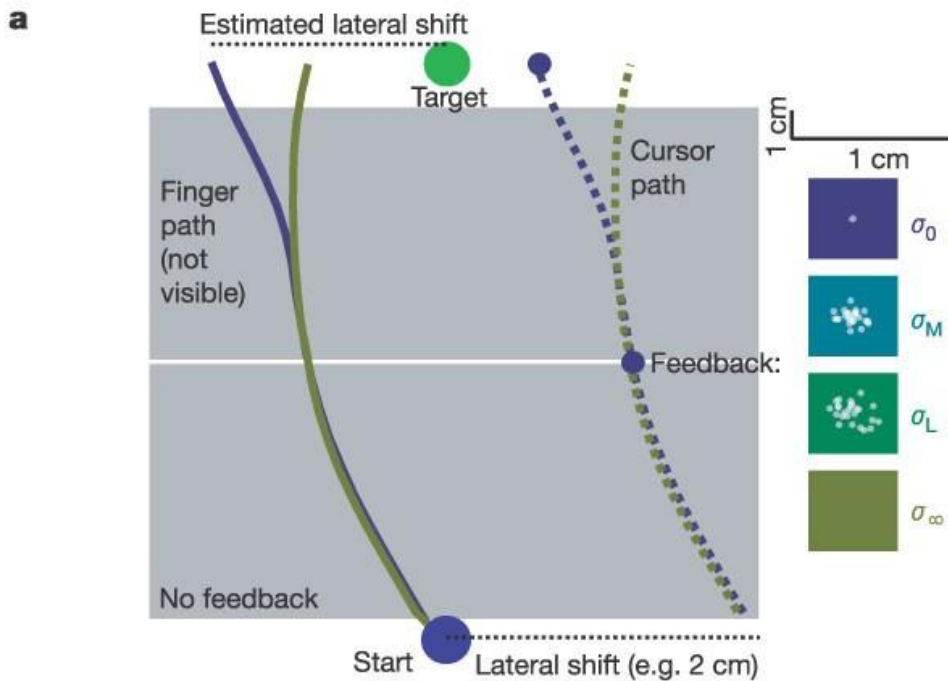
Sobell Department of Motor Neuroscience, Institute of Neurology,
University College London, Queen Square, London WC1N 3BG, UK

When we learn a new motor skill, such as playing an approaching tennis ball, both our sensors and the task possess variability. Our sensors provide imperfect information about the ball's velocity, so we can only estimate it. Combining information from multiple modalities can reduce the error in this estimate¹⁻⁴. On a longer time scale, not all velocities are a priori equally probable, and over the course of a match there will be a probability distribution of velocities. According to bayesian theory^{5,6}, an optimal estimate results from combining information about the distribution of velocities—the prior—with evidence from sensory feedback. As uncertainty increases, when playing in fog or at dusk, the system should increasingly rely on prior knowledge. To use a bayesian strategy, the brain would need to represent the prior distribution and the level of uncertainty in the sensory feedback. Here we control the statistical variations of a new sensorimotor task and manipulate the uncertainty of the sensory feedback. We show that subjects internally represent both the statistical distribution of the task and their sensory uncertainty, combining them in a manner consistent with a performance-optimizing bayesian process^{4,5}. The central nervous system therefore employs probabilistic models during sensorimotor learning.

There are several possible computational models that subjects could use to determine the compensation needed to reach the target on the basis of the sensed location of the finger midway through the movement. First (model 1), subjects could compensate fully for the



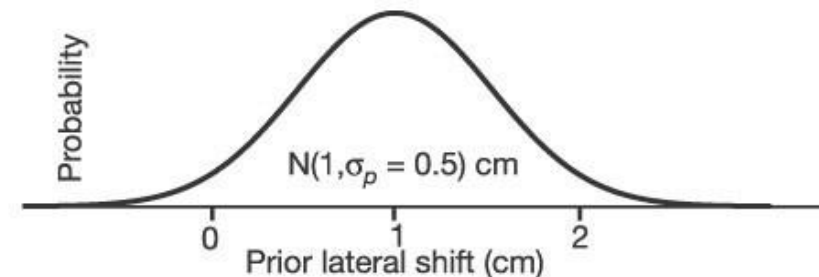
Sensory Cue Integration



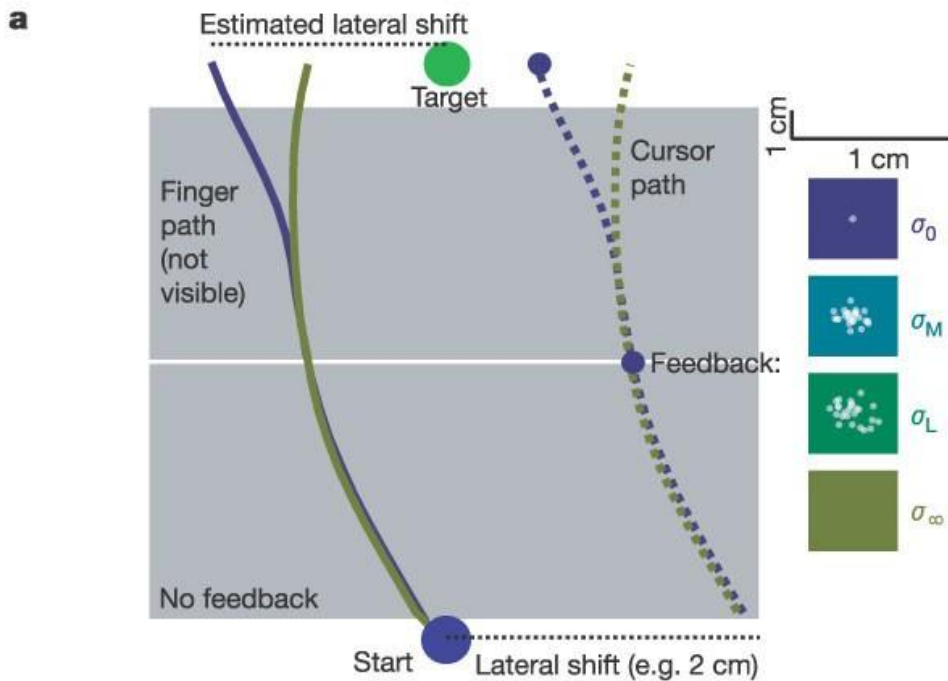
- Subject has to reach a visual target with index finger
- Lateral shift is introduced
- Trained for 1000 trials

→ Experience and **prior** knowledge

$$p_{\text{prior}} = \mathcal{N}(\mu = 1\text{cm}, \sigma = 0.5\text{cm})$$



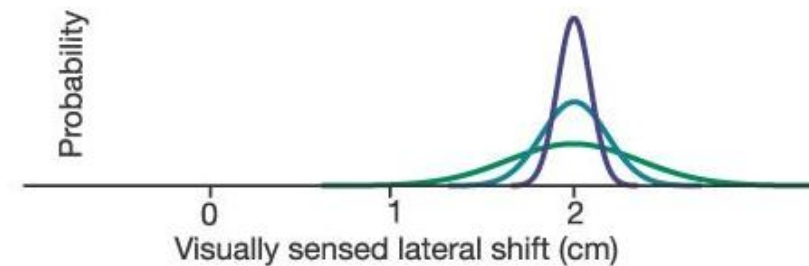
Sensory Cue Integration



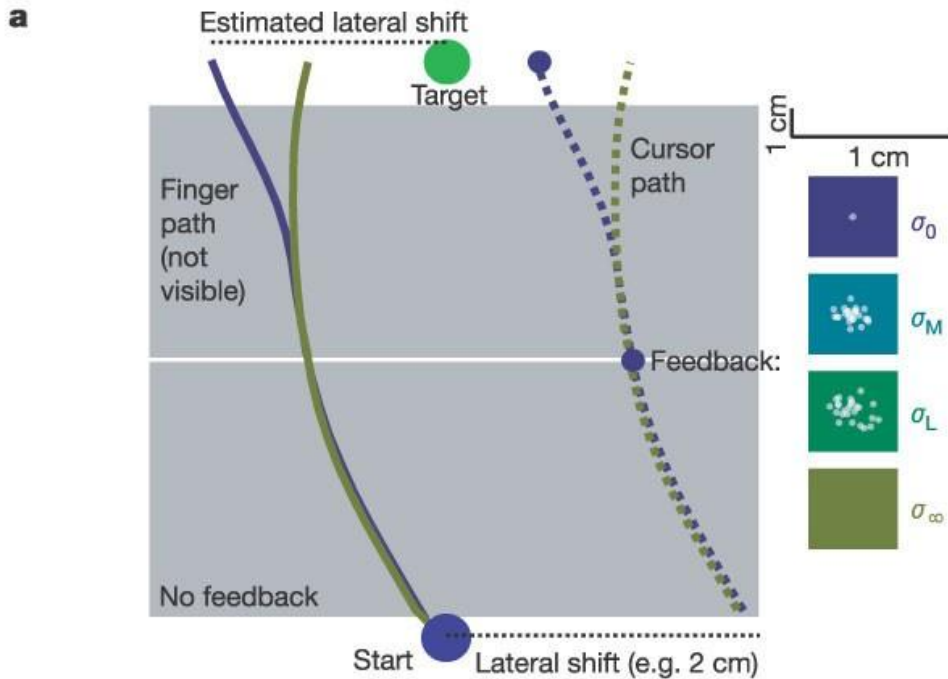
- Visual feedback with uncertainty σ_0 , σ_M and σ_L

→ Sensory information and **evidence**

$$p_{evidence} = p(x_{sensed} | x_{true})$$



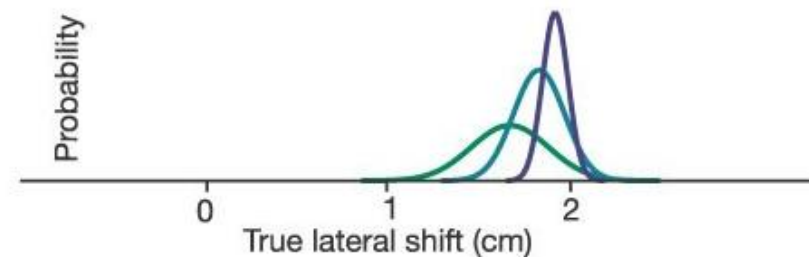
Sensory Cue Integration



- estimate target position

→ Combining **prior** knowledge and **evidence** (Bayes' theorem)

$$p_{\text{posterior}} = \frac{p_{\text{evidence}}(x_{\text{sensed}} | x_{\text{true}}) p_{\text{prior}}(x_{\text{true}})}{p(x_{\text{sensed}})}$$



Summary

1. Recap
 - Neuron models
2. Learning
 - Learning Models
 - Hebbian Learning
 - Oja's Rule
 - BCM Theory
 - STDP
3. Conversion of ANNs to SNNs
 - SpikeProp
 - Activation Functions
 - CNN
 - RNN
4. Learning over years
 - Neuroplasticity
 - Epigenetic Changes
5. Sensory Information and Learning
 - Simple Cells
 - Bayesian Sensory Integration